MIL-T-82330A
11 August 1969
SUPERSEDING
MIL-T-82330(ONR/NTDC)
22 September 1965

MILITARY SPECIFICATION

TRAINER, BASIC ELECTRONICS, STUDENT

This specification is mandatory for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for the construction of the Basic Electronics Student Trainer (BEST)

1.1.1 This item has been designated by the Naval Training Device Center as Device 6B19 for Navy procurement (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, wood, cleated plywood
PPP-B-636	Box, Fiberboard
PPP-T-60	Tape, pressure sensitive adhesive, waterproof, for packaging

Military

MIL-P-116	Preservation; Methods of
MIL-E-16400	Electronic Equipment, Naval Ship and Shorte; General Specification for
MIL-L-10547	Liners, case and sheet, overwraps, water-vaporproof, flexible

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MIL-T-23991	Training Devices, Military; General Specifications For

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-440	Soldering Techniques for Standard Type Solder Terminals

DRAWINGS

Naval Training Device Center (NAVTRADEVCEN)

IL 100640	Index List for Device 6B19

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be as specified on the drawings of IL 100640. Materials not specified in IL 100640 shall be selected in accordance with MIL-T-23991.

3.2 Design.- The design of the trainer shall conform to the drawings of IL 100640.

3.2.1 Interchangeability.- All parts, subassemblies, and assemblies having the same part number shall be directly and completely interchangeable with each other, in respect to installation and performance, regardless of sources from which obtained, without alteration and without selection for fit or performance.

3.2.2 Safety.- With the exception of the interlocks requirements, components shall be installed to comply with the safety requirements of MIL-E-16400.

3.2.2.1 Grounding cable.- The basic power and master power units shall be equipped with a 10-foot, 3-wire power cable. The cable shall be mechanically secured in the units to prevent stress to the soldered connection. The ground wire shall not be interrupted by power switching devices.

3.3 Construction.- Construction shall be in accordance with the drawings of IL 100640 and 3.3.1 through 3.3.3.

3.3.1 Component mounting boards.- Component mounting boards shall be of 3/32-inch melamine.

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3.3.2 Soldering.- Soldering techniques shall comply with MIL-STD-440.

3.3.3 Wire color coding.- Differentiation color coding of chassis wiring shall be used for identification of circuits and checkpoints to comply with IL 100640.

3.4 Performance.-

3.4.1 The trainer shall be capable of operation, by using the components listed in 3.5, as (a) a single subchassis, (b) a combination of subchassis, or (c) a specific system, as required for training purposes. Each plug-in subchassis shall be capable of being unplugged from its power receptacle plug unit, rotated 180°, and replugged to assume a back-to-front position for the purpose of gaining access to the subchassis' rear connections.

3.4.2 Controls and components.- All controls and movable components shall operate smoothly without binding. The controls and components, when set to a selected position, shall not reset to a different position until moved manually.

3.5 Details of components.- The trainer shall consist of the following major units and subchassis, the details of which are shown in Drawings of IL 100640:

(a) Units	NAVTRADEVCEN No.	NAVPERS No.
(1) Basic Power Unit	6B19/1	17540-1
(2) Master Power Unit	6B19/29	-29
(3) Plug Unit (3-plug)	6B19/27	-27
(4) Plug Unit (6-plug)	6B19/28	-28
(b) Subchassis:		
(1) Voltage Regulator	6B19/2	17540A-2
(2) RF Amplifier	6B19/3	-3
(3) Oscillator R.F.	6B19/4	-4
(4) Mixer-Converter	6B19/5	-5
(5) IF Amplifier (455 KC)	6B19/6	-6
(6) Detector	6B19/7	-7
(7) Audio Amplifier	6B19/8	-8
(8) Transistor Analyzer	6B19/9	-9

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(9) Transistor Analyzer	6B19/10	17540A-10
(10) Transistor Receiver	6B19/11	-11
(11) Relay Line	6B19/12	-12
(12) RC Oscillator	6B19/13	-13
(13) Klystron Transceiver	6B19/14	-14
(14) IF Amplifier (Video)	6B19/15	-15
(15) Cathode Follower	6B19/16	-16
(16) Video Amplifier	6B19/17	-17
(17) Modulator (FM-AM)	6B19/18	-18
(18) Wave Shaper	6B19/19	-19
(19) Clipper-Clamper	6B19/20	-20
(20) Sweep Generator	6B19/21	-21
(21) Multi-Vibrator	6B19/22	-22
(22) FM Detector	6B19/23	-23
(23) Magnetic Amplifier	6B19/24	-24
(24) Counter	6B19/25	-25
(25) Class "B" and "C" Amplifier	6B19/26	-26

3.6 Dimensions and tolerances.- Dimensions and tolerances shall be as shown in Drawings of IL 100640. Tolerances not shown shall be maintained to assure meeting the interchangeability and performance requirements specified in 3.2.1 and 3.4 respectively. Electrical tolerances shall be $\pm$ 10 percent.

3.7 Finish and color.- Finish and color shall be in accordance with the drawings of IL 100640.

3.8 Product markings.- NAVTRADEVCEN and NAVPERS identification numbers as shown in 3.5 shall be in accordance with the drawings of IL 100640.

3.9 Workmanship.- The trainer shall be constructed and finished with particular attention given to neatness and thoroughness of subchassis construction, etching, silk screening, sanding, grinding, riveting, machine screw assemblage, anodizing, wiring, soldering, markings of parts and assemblies, and freedon from sharp edges and burrs.

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3.9.1 Cleanup.- Units shall be thoroughly cleaned of loose, splattered or excess solder, metal chips, and other foreign material after final assembly.

3.9.2 Wire Stripping.- Wire stripping shall be done without nicking or otherwise damaging the wire.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspections to be performed are classified as follows:

(a) In-process inspection

(b) Quality conformance inspection

4.3 In-process inspection.- In-process inspection shall include such visual, electrical, and mechanical examinations and testing of materials, subassemblies, parts, and accessories (including purchased standard items) during the manufacturing process as may be necessary to assure conformance to all the requirements specified herein.

4.4 Quality conformance inspection.-

4.4.1 Preproduction inspection.- Preproduction inspection shall be conducted on the preproduction sample and shall consist of all the examinations and tests specified in 4.5 and 4.6.

4.4.2 Production inspection.- Production units shall be inspected to the extent necessary to maintain quality control and conformance to the Government accepted preproduction sample.

4.5 Examinations.- Examinations shall be performed to verify conformance to the following requirements:

(a) Materials

(b) Construction

(c) Safety

(d) Dimensions and tolerances

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(e) Finish and color

(f) Product markings

(g) Workmanship

(h) Preparation for delivery

4.6 Tests.- Electrical operating tests specified in drawings of IL 100640 shall be performed using the test equipment or equivalent specified therein. Test equipment will be certified by the Government inspector. Results of tests shall be recorded and furnished with each unit delivered.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be Level A or C (see 6.2).

5.1.1 Level A.- Each trainer shall be preserved and packaged in accordance with MIL-P-116, Method II.

5.1.2 Level C.- Trainers shall be preserved and packaged to a degree which will afford protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity. This level, in general, will conform to applicable carrier rules and regulations. Suppliers commercial practices may be used when they meet the requirements of this level.

5.2 Packing.- Packing shall be Level A, B, or C (see 6.2).

5.2.1 Level A.- Units preserved and packaged in accordance with 5.1.1 shall be packed in exterior type shipping containers conforming to PPP-B-601. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. Containers shall be provided with a case liner conforming to MIL-L-10547 and shall be sealed in accordance with the appendix thereto.

5.2.2 Level B.- Units preserved and packaged in accordance with 5.1.1 shall be packed in exterior type shipping containers conforming to PPP-B-636. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. Containers shall be sealed with tape conforming to PPP-T-60 in accordance with the appendix thereto.

5.2.3 Level C.- Packages which require over-packing for acceptance by the carrier, shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet consolidated freight classification rules and regulations of other common carriers as applicable to the mode of transportation.

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5.3 Physical protection.- All material shall be packaged and packed in containers constructed to provide a compact and nonshifting load. Adequate cushioning, blocking and bracing shall be provided within shipping containers in a manner designed to prevent damage due to shcok, vibration or impact during shipment, handling and storage.

5.4 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The trainer will be used in a classroom for basic electronics student training in conjunction with a course curriculum.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Procuring activity designation (device number or other) (see 1.1.1)

(c) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

6.3 Definitions.- Except for the following, definitions will be furnished upon written request to the procuring activity.

6.3.1 Preproduction sample.- A production item furnished under production contracts for the purpose of determining compliance with the requirements of the detail training device specification already established and accepted. It shall be electrically and mechanically representative of production sets though not necessarily made with production line jigs, fixtures and methods.

6.3.2 Production unit.- A unit in its final mechanical and electrical form of final production design made by production tools, jigs, fixtures and methods.

CUSTODIAN
Navy-TD
Army-EL
Air Force-70

PREPARING ACTIVITY:
Navy-TD
Project No. 6940-0024

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-T-82330A

ORGANIZATION *(Of submitter)*
NAVAL TRAINING DEVICE CENTER

CITY AND STATE
ORLANDO, FLA 32813

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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Naval Training Device Center
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